EASY ALGEBRA

Systems of two linear equations in two variables

01

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 GRID-IN ALGEBRA

$$3x + y = 29$$
$$x = 2$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ALGEBRA

An online bookstore sells novels and magazines. Each novel sells for $4, and each magazine sells for $1. If Sadie purchased a total of 11 novels and magazines that have a combined selling price of $20, how many novels did she purchase?

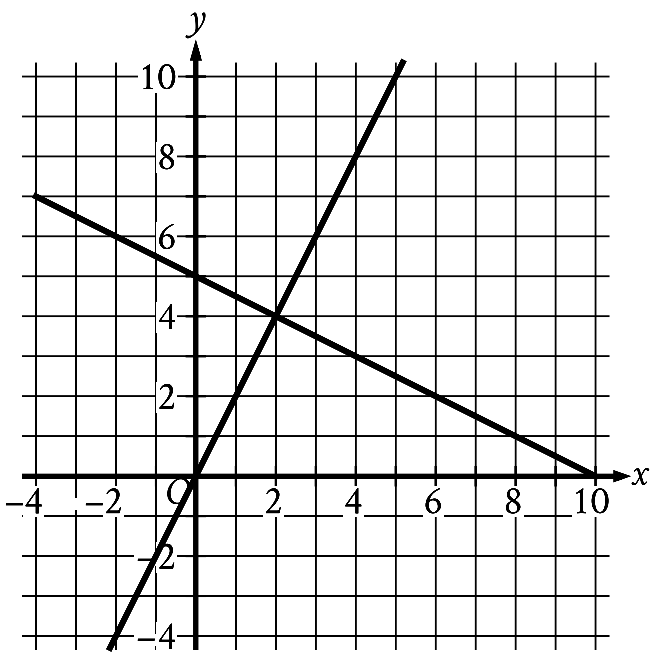
A. 2

B. 3

C. 4

D. 5

Q3 ALGEBRA



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma 5)
    - (2 comma 4)
    - (10 comma 0)
- For the second line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 0)
    - (2 comma 4)
    - (5 comma 10)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. $(0, 5)$

B. $(2, 4)$

C. $(5, 10)$

D. $(10, 0)$

Q4

ALGEBRA

A movie theater charges $11 for each full-price ticket and $8.25 for each reduced-price ticket. For one movie showing, the theater sold a total of 214 full-price and reduced-price tickets for $2,145. Which of the following systems of equations could be used to determine the number of full-price tickets, f, and the number of reduced-price tickets, r, sold?

A. $f + r = 2{,}145$
$11f + 8.25r = 214$

B. $f + r = 214$
$11f + 8.25r = 2{,}145$

C. $f + r = 214$
$8.25f + 11r = 2{,}145$

D. $f + r = 2{,}145$
$8.25f + 11r = 214$

Q5 ALGEBRA

$$x + y = 18$$

$$5y = x$$

What is the solution x, y to the given system of equations?

A. $15, 3$

B. $16, 2$

C. $17, 1$

D. $18, 0$

Q6

GRID-IN ALGEBRA

$$x + y = 125$$

$$x + y + y = 155$$

The solution to the given system of equations is x, y. What is the value of y?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

ALGEBRA

Angela is playing a video game. In this game, players can score points only by collecting coins and stars. Each coin is worth c points, and each star is worth s points.

- The first time she played, Angela scored 700 points. She collected 20 coins and 10 stars.
- The second time she played, Angela scored 850 points. She collected 25 coins and 12 stars.

Which system of equations can be used to correctly determine the values of c and s ?

A. $10c + 20s = 700$
$12c + 25s = 850$

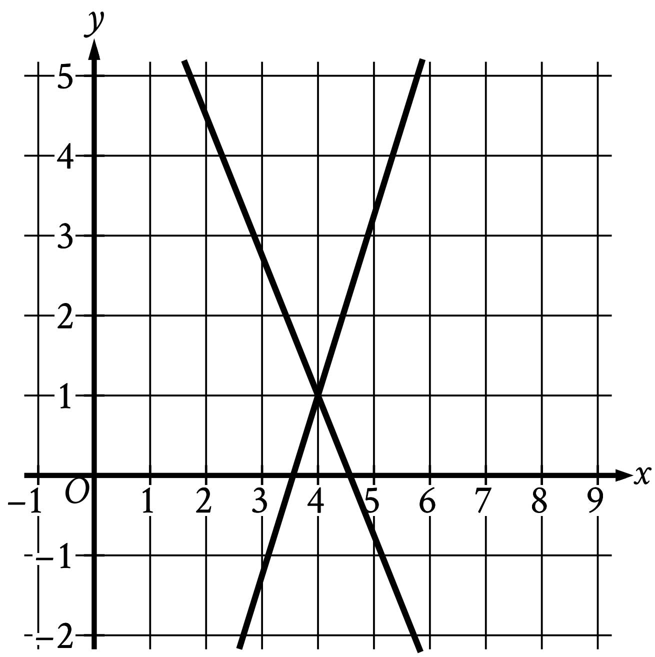
B. $20c + 10s = 700$
$25c + 12s = 850$

C. $20c + 700s = 10$
$25c + 850s = 12$

D. $700c + 20s = 10$
$850c + 25s = 12$

Q8

GRID-IN ALGEBRA



- For the first line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (4 comma 1)
    - (StartFraction 32 Over 7 EndFraction comma 0)
- For the second line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (StartFraction 32 Over 9 EndFraction comma 0)
    - (4 comma 1)

The graph of a system of linear equations is shown. The solution to the system is x, y. What is the value of x?

STUDENT-PRODUCED RESPONSE

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 ALGEBRA

$$2x + 7y = 9$$
$$8x + 28y = a$$

In the given system of equations, a is a constant. If the system has infinitely many solutions, what is the value of a ?

A. 4

B. 9

C. 36

D. 54

Q10 ALGEBRA

$$y = -3x$$

$$4x + y = 15$$

The solution to the given system of equations is x, y. What is the value of x?

A. 1

B. 5

C. 15

D. 45